Classical Optimization Algorithms for Lasso Regression

A Review and Empirical Comparison

I Background | II Algorithms | III Results

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PART I
Background



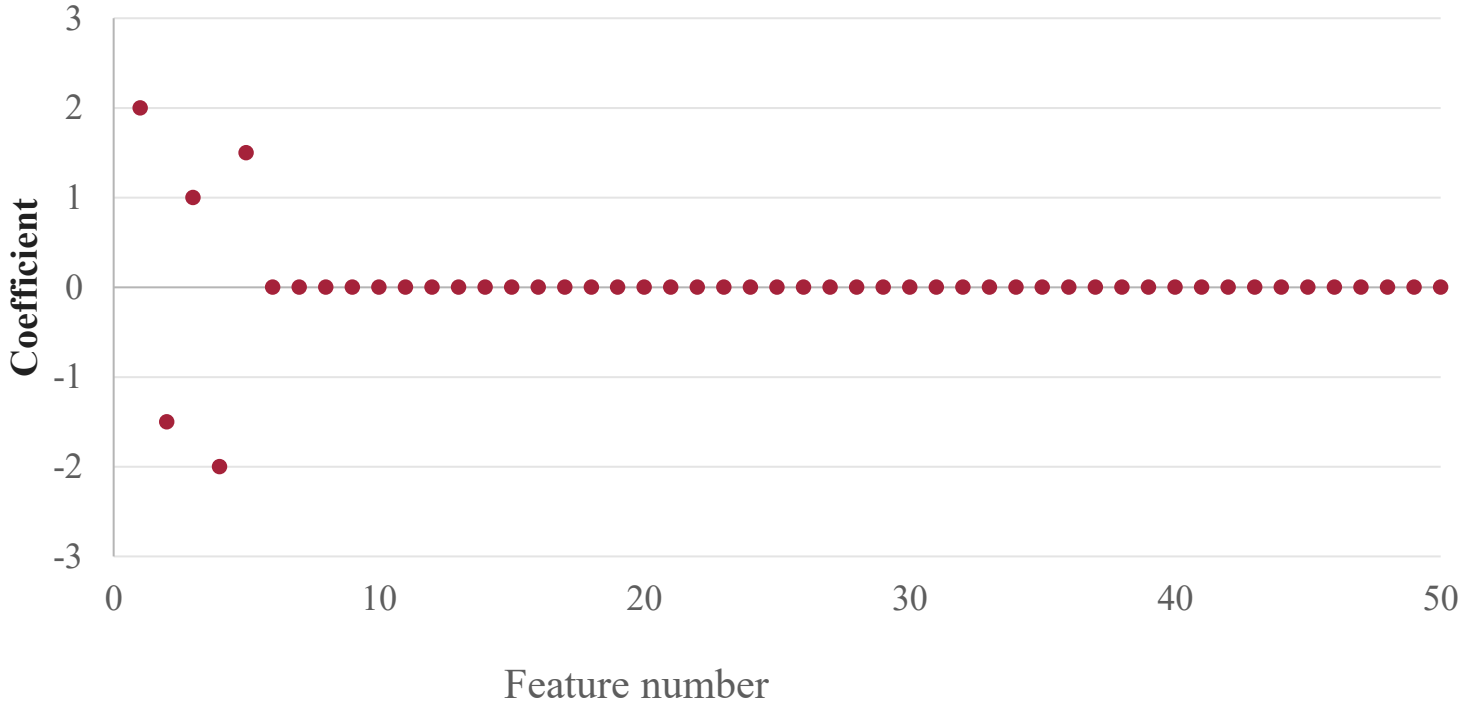
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Why study Lasso?

Lasso combines variable selection with a convex, nonsmooth objective.

Known coefficients in our synthetic data



50

candidate features

5

generating variables

Zero coefficients simplify the fitted linear model.

Sparse modeling provides a clear setting for comparing optimization methods.

Source: Group 9 report, Sections 1, 2.4 and 4.1. Chart shows the generating coefficients, not estimated coefficients.

The Lasso training problem

The same training objective underlies every solver in our comparison.

$$\text{minimize} \quad F(w) = g(w) + \lambda \|w\|_1$$

over $w \in \mathbb{R}^d$

Squared training loss

$$g(w) = \|Xw - y\|_2^2 / (2n)$$

Measures prediction error on the training data.

L1 regularization

$$\lambda \|w\|_1 = \lambda \, \Sigma_j \, |w_j|$$

λ controls regularization. Larger λ strengthens the penalty.

X: $n \times d$ training features y: centered responses w: coefficients

Use training statistics for centering and scaling. Keep the intercept unpenalized.

Source: Group 9 report, Section 2.1, Equation 1. Tibshirani (1996).

Related regression models

The same squared loss can be paired with different penalties.

Model	Penalty added to g(w)	Typical behavior
Ridge	$(\alpha/2)\|w\|_2^2$	Shrinks coefficients Usually retains a dense model
Lasso	$\lambda\|w\|_1$	Shrinks coefficients Can produce exact zeros
Elastic Net	$\lambda\|w\|_1 + (\alpha/2)\|w\|_2^2$	Encourages grouped selection of correlated predictors

$\alpha, \lambda > 0$. The loss is the same. The penalty changes the model.

Our experiments compare solvers for a fixed Lasso objective.

Source: Group 9 report, Section 2.5, Table 2. Hoerl and Kennard (1970), Tibshirani (1996), Zou and Hastie (2005).

Convexity and sparse optimality

Convexity supports global optimization. The L1 penalty permits exact zeros.

Convexity

Squared loss and the L1 norm are convex.
Every local minimum is global.

Constrained form: minimize g(w)
subject to $\|w\|_1 \le t$

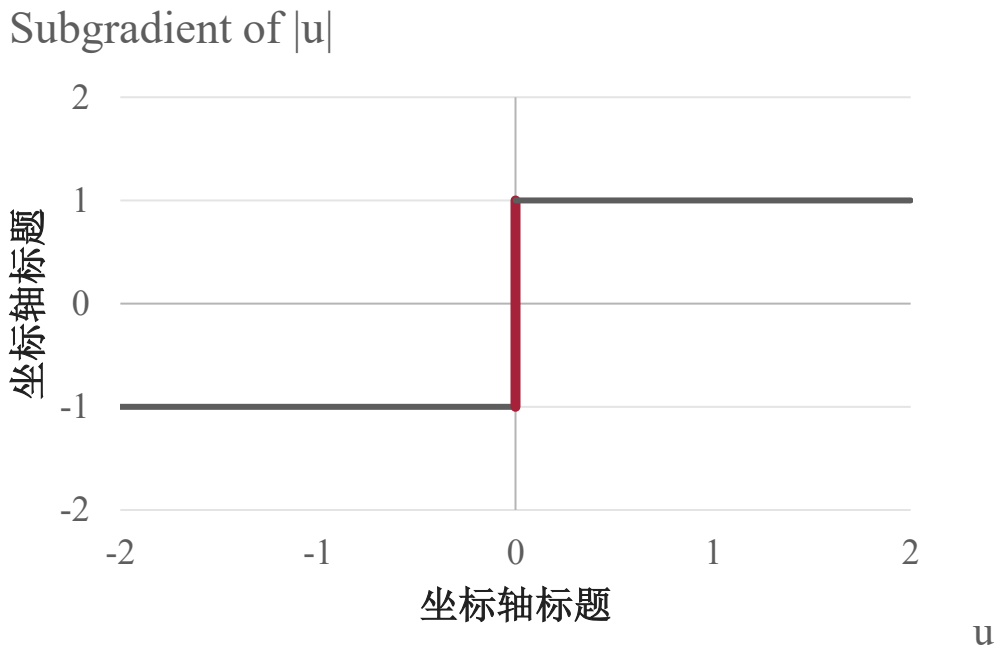
For a penalized optimum, choose $t = \|w^*\|_1$.

Optimality at w*

$$0 \in \nabla g(w^*) + \lambda \partial \|w^*\|_1$$

$$w^*_j = 0: \ |\nabla_j g(w^*)| \le \lambda$$

A gradient interval permits exact zeros.



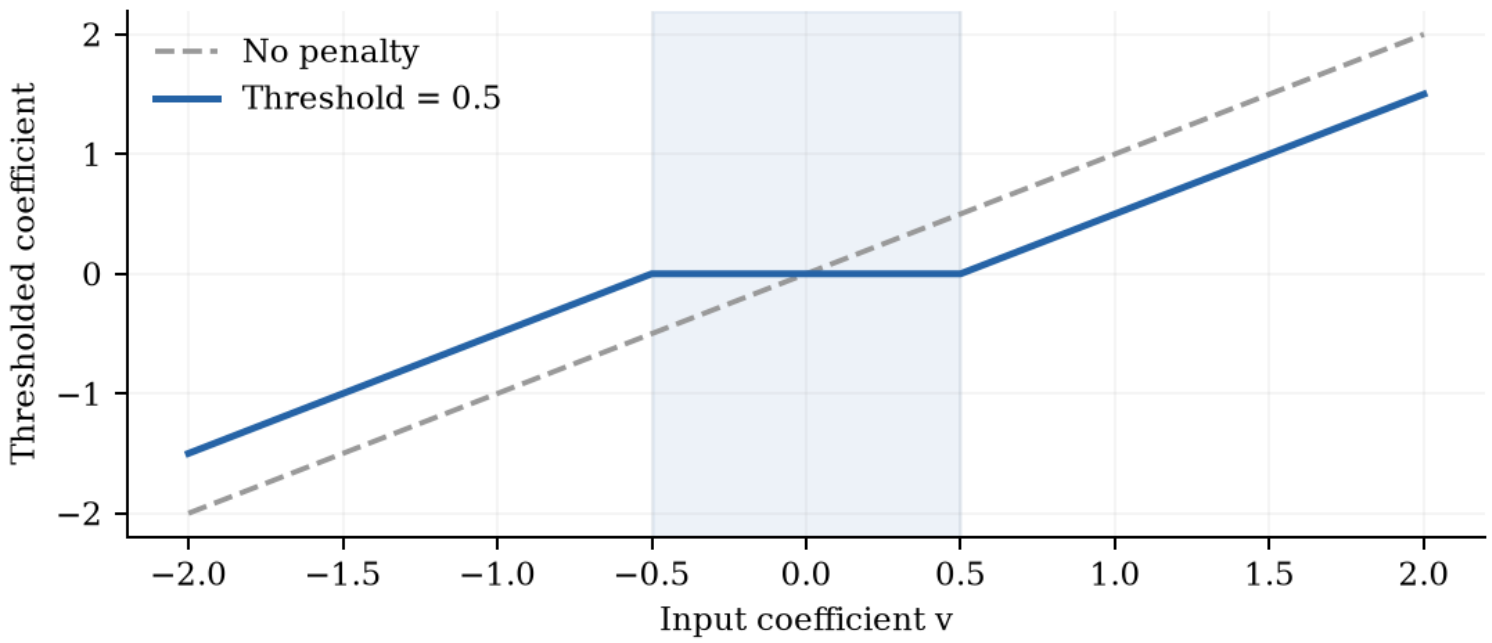
$$\partial|0| = [-1, 1]$$

Source: Group 9 report, Sections 2.2–2.4. Optimality conditions apply jointly at w*.

Soft thresholding

Small inputs become zero. Larger inputs shrink toward zero.

$$S_\tau(v) = \text{sign}(v)\,\max(|v| - \tau, 0)$$



Example with τ = 0.5

Input v	$S_{0.5}(v)$
1.2	0.7
−0.8	−0.3
0.2	0

Scalar proximal solution: $\arg\min_u \tfrac{1}{2}(u - v)^2 + \tau|u|$

Source: Group 9 report, Section 2.4, Equation 6 and Figure 1. Original package figure. Threshold τ = 0.5.

Scope and comparison question

Scope: convex squared-loss Lasso for linear regression.

Same data, preprocessing and λ

ISTA

Gradient step
and thresholding

Daubechies et al. (2004)

FISTA

Extrapolation
and thresholding

Beck & Teboulle (2009)

Cyclic CD

Exact minimization
of one coordinate

Friedman et al. (2010)

How much computation is needed to reach a common accuracy target?

We examine how feature relationships and dimension change the cost.

Next: algorithm updates and the common experimental protocol.

Sources: Group 9 report, Sections 1.1 and 3. Daubechies et al. (2004), Beck and Teboulle (2009), Friedman et al. (2010).

PART II
Algorithms

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Three Solvers, One Lasso Objective

Common problem

$$\min_{w} F(w) = \frac{1}{2n}\|Xw - y\|_2^2 + \lambda\|w\|_1$$

Method	What is updated?	Key idea
ISTA	Full coefficient vector	Gradient + soft thresholding
FISTA	Full vector at extrapolated point	Add momentum
Cyclic CD	One coefficient at a time	Exact scalar minimization

Key comparison

All three use **soft thresholding**; they differ in *where* and *how* it is applied.

ISTA vs. FISTA: Proximal Gradient Updates

Soft-thresholding operator

$$S_\tau(v)_j = \text{sign}(v_j)\max\{|v_j| - \tau, 0\}$$

ISTA: current iterate

$$w^{k+1} = S_{\eta\lambda}\left(w^k - \eta\nabla g(w^k)\right)$$

- Simple; objective nonincreasing
- Objective gap: $O(1/k)$

FISTA: extrapolated iterate

$$w^k = S_{\eta\lambda}\left(z^k - \eta\nabla g(z^k)\right)$$

$$z^{k+1} = w^k + \frac{t_k - 1}{t_{k+1}}(w^k - w^{k-1})$$

- Accelerated worst-case bound: $O(1/k^2)$
- Standard version need not be monotone

Important distinction

A faster *objective-error bound* does not guarantee lower runtime or faster stationarity convergence.

Cyclic Coordinate Descent: One Variable at a Time

- Fix all coefficients except w_j and solve a one-dimensional Lasso problem.
- Use the latest partial residual $r^{(j)} = r + x_j w_j$, where $r = y - Xw$.

Exact coordinate update

$$w_j \leftarrow \frac{S_\lambda\left(x_j^\top r^{(j)}/n\right)}{\|x_j\|_2^2/n}$$

- Update the residual after each coordinate; a complete scan is one **sweep**.
- No global gradient step size is needed; objective is nonincreasing.
- Dense work: $O(n)$ per coordinate, $O(nd)$ per sweep.

Fair counting

One CD sweep includes d scalar updates—it is not the same operation as one ISTA/FISTA round.

Beyond the Three Solvers

Family	Main mechanism	Additional work / trade-off
LARS–Lasso	Trace solution path as λ changes	Active-set linear algebra
Interior point	Barrier / Newton steps	Linear-system solves
ADMM	Split loss and penalty	Linear solve + dual state
Restarted FISTA	Reset unhelpful momentum	Restart decision
Safe screening	Certify coefficients are zero	Screening certificates

Why not benchmark them all?

Our experiments isolate **three basic fixed-penalty solvers**; path construction and advanced methods involve different setup and computational costs.

Strengths, Weaknesses, and Fair Comparison

Property	ISTA	FISTA	Cyclic CD
Objective monotone	Yes	Not guaranteed	Yes
General objective-gap bound	$O(1/k)$	$O(1/k^2)$	No universal rate stated
Step size $1/L$ needed	Yes	Yes	No
Extra state	Minimal	Momentum z, t	Residual r

- Same asymptotic dense work per round, $O(nd)$, but **different implementation overhead**.
- Compare **elapsed time and a common stopping criterion**, not round counts alone.
- Keep λ fixed for solver comparisons: changing it changes the *model*.

Takeaway

Theory describes objective-error guarantees; experiments measure practical cost to a specified tolerance.

Datasets and Common Experimental Settings

Datasets

Dataset	Instances	(N, d)	Train/test
Synthetic	3	$(300, 50)$	240/60
Diabetes	1	$(442, 10)$	353/89

- Synthetic: 5 true nonzeros; Gaussian noise ($\sigma = 0.5$)
- Diabetes: real regression data; true support unknown

Shared settings

- Train-only centering and scaling
- Zero initialization: $w^0 = 0$
- $\lambda = 0.1\lambda_{\max}$, $\lambda_{\max} = \|X^\top y/n\|_\infty$
- ISTA/FISTA: $\eta = 1/L$
- Cyclic CD: $j = 1, \ldots, d$
- Maximum: 10^4 rounds

Controlled design

All solvers receive the **same preprocessed inputs, penalty, and data splits.**

Normalized proximal-gradient residual

$$G_L(w) = L\left[w - S_{\lambda/L}\left(w - \frac{\nabla g(w)}{L}\right)\right] \quad R(w) = \frac{\|G_L(w)\|_\infty}{\max\{1, \|X^\top y/n\|_\infty\}} \le 10^{-6}$$

- Evaluate at the **returned coefficients** after every vector update / CD sweep.
- Measure objective, residual, rounds, **elapsed time**, test MSE, and nonzero coefficients.
- Five timing repetitions; report **median** per instance.
- Validate updates against orthogonal soft thresholding and a tighter scikit-learn solution.

Transition to results

This protocol separates **optimization efficiency** from **prediction and sparsity**—the next section compares the measured outcomes.

PART III
Results

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The same Lasso objective gives nearly identical fitted results.

Test MSE at the main penalty

Method	Synthetic	Diabetes
ISTA	0.395983155	2966.935452
FISTA	0.395981893	2966.936439
CD	0.395982019	2966.936609

Synthetic: mean over 3 instances. Diabetes: one fixed split.

$R \le 10^{-6}$

All main runs meet the stopping threshold.

5

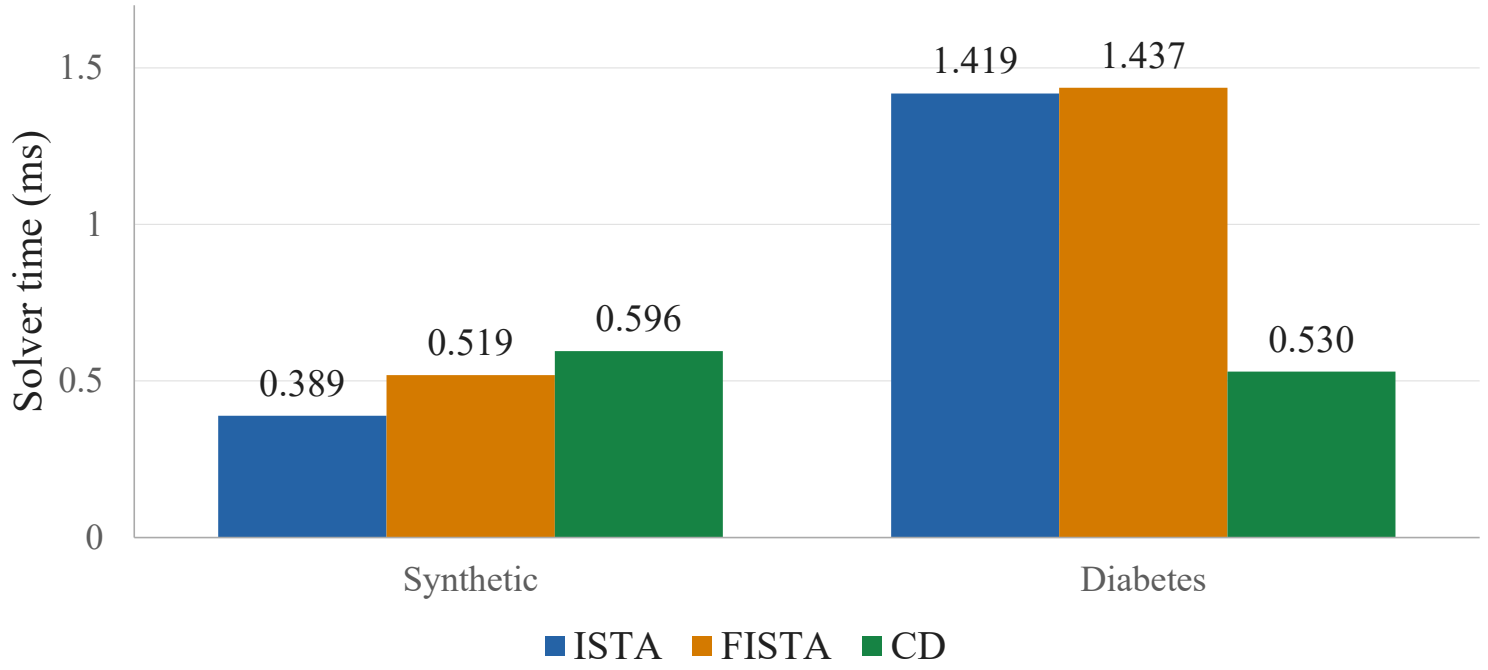
nonzero coefficients for every method and instance

Comparable predictions let us focus on the cost of solving the same problem.

Source: Group 9 report, Sections 5.1 and 5.4. Main penalty: $\lambda/\lambda_{max} = 0.1$.

Runtime depends on the dataset

The fastest solver changes between synthetic data and Diabetes.



Synthetic: mean of 3 instance medians. Diabetes: median of 5 timings.

ISTA is fastest on Synthetic. CD is fastest on the fixed Diabetes split.

Source: original benchmark CSVs, 2026-10-06. Solver time includes monitoring and excludes shared spectral setup.

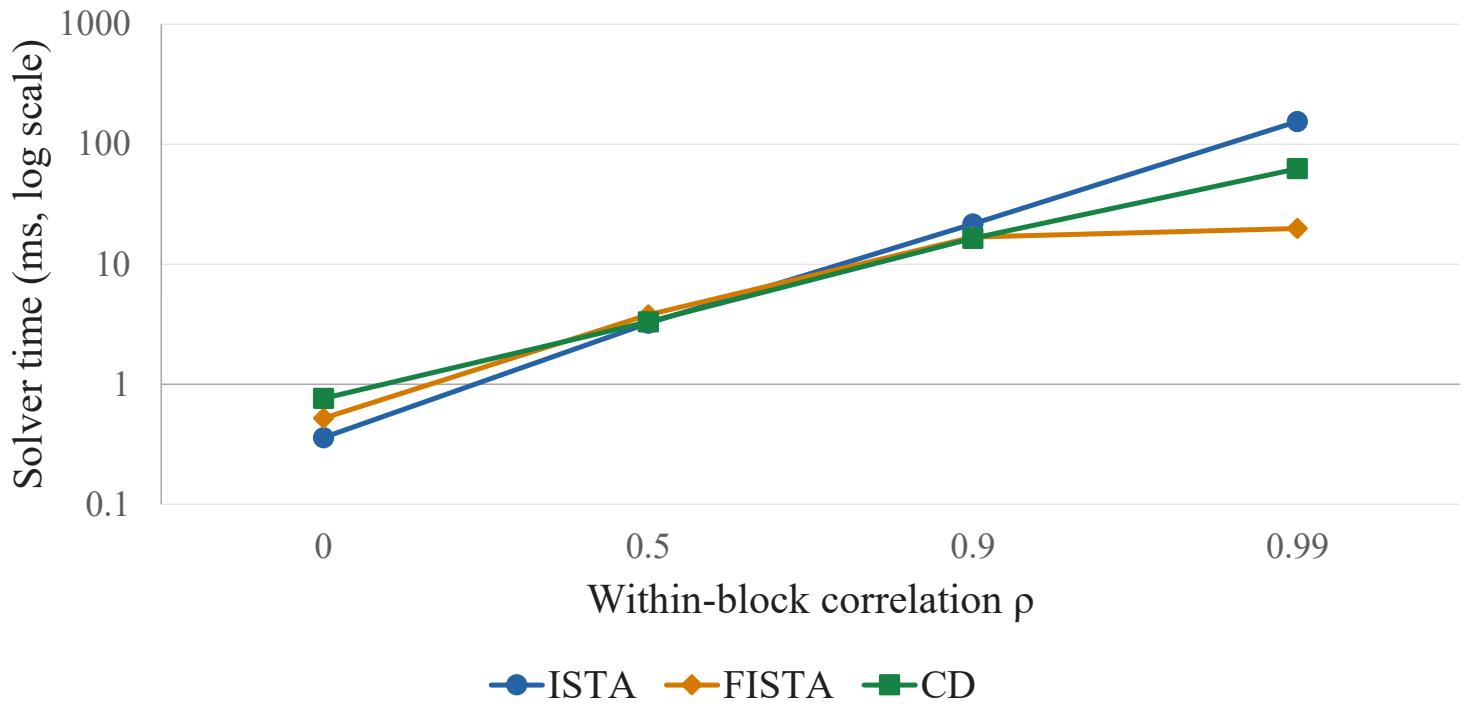
Why FISTA can take longer

Its theory bounds objective error.

Our stopping rule uses a residual, which can fluctuate.

Strong correlation changes the ranking

FISTA has the lowest mean time in the strongest correlation setting.



At ρ = 0.99

Method	Time (ms)
ISTA	154.6 ± 10.5
FISTA	19.9 ± 18.2
CD	62.5 ± 8.7

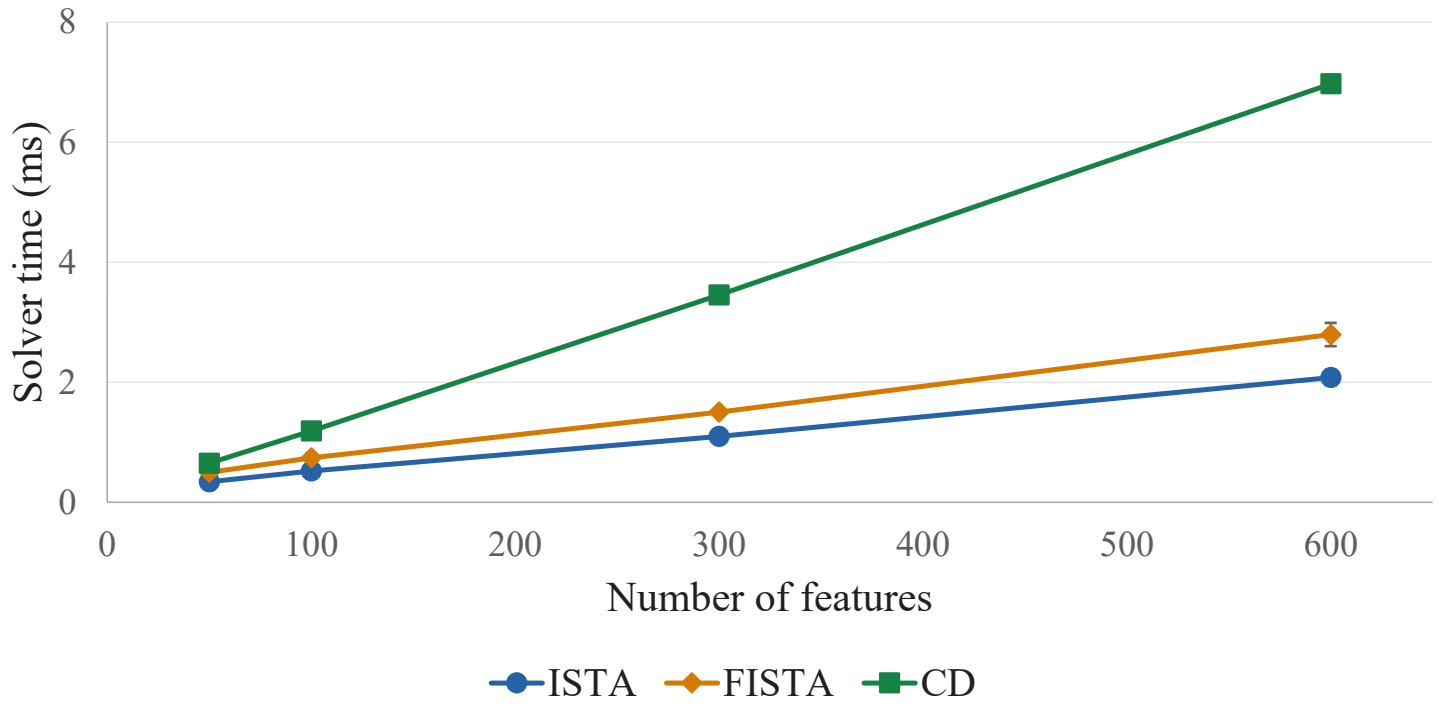
FISTA has substantial variation across instances.

Acceleration helps in this difficult setting. The ranking changes with the data.

Source: extended aggregated results, 2026-10-07. Error bars: SD across three instance medians, five timings each.

A CD sweep gets costlier as features grow

A complete sweep updates every coefficient once.



Fixed training size: 800 samples

4 complete sweeps

for CD at every tested size

Features	Coordinate updates
50	200
600	2,400

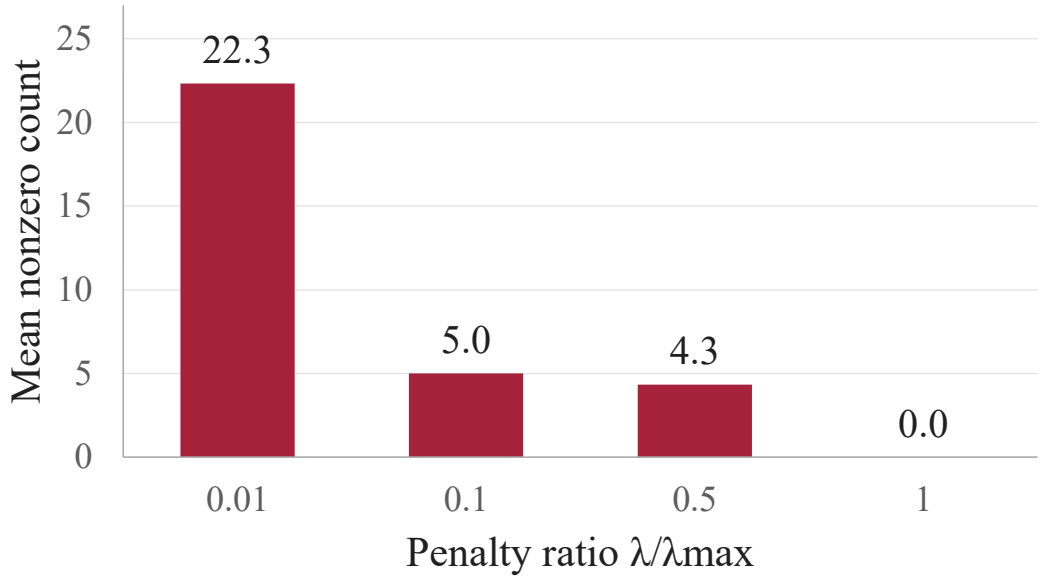
A small sweep count can still involve substantial work.

Source: extended aggregated results, 2026-10-07. Independent features, 800 training samples. Error bars: sample SD.

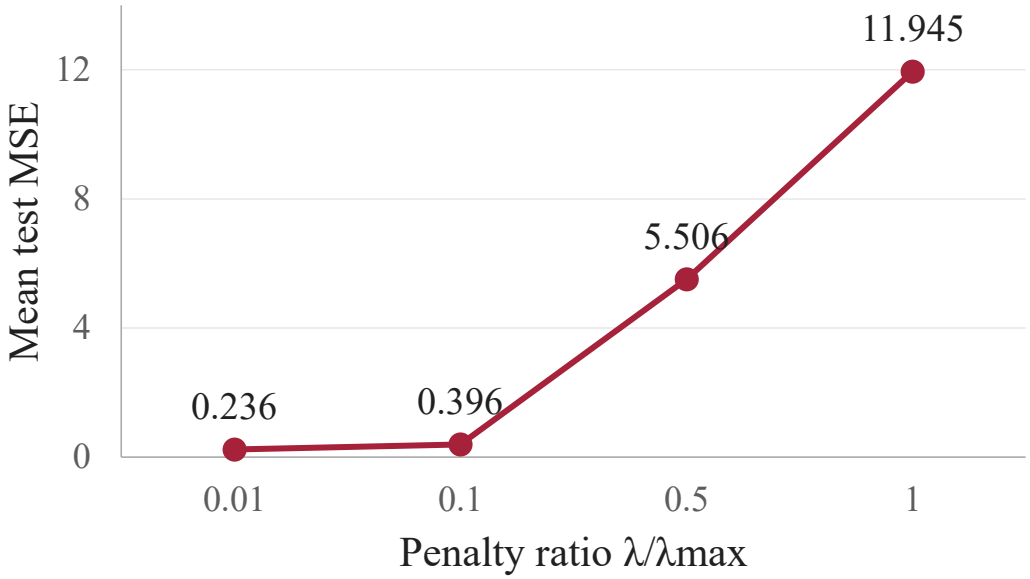
Sparser models can have larger test errors

A stronger penalty trades a smaller model against fitting error.

Selected variables



Test prediction error



At ratio 0.1, the model selects the 5 generating variables, but still shrinks their coefficients.

Fewer variables do not guarantee a smaller prediction error.

Source: regularization_results.csv, 2026-10-06. Means over three synthetic instances, solved using CD.

Conclusions and limitations

The best solver depends on the data and the implementation.

Method	Observed advantage in our experiments
ISTA	Lowest time on the tested independent-feature problems
FISTA	Lowest mean time at the strongest correlation, $\rho = 0.99$
Cyclic CD	Lowest time on the fixed Diabetes split

All three methods produce nearly identical fitted results after convergence.

Stronger regularization produces a smaller model and can increase test error.

Scope of the evidence

Three synthetic instances per setting, one Diabetes split, and the current implementations.

Source: Group 9 report, Section 6. Findings apply to the recorded datasets, stopping rule and implementations.